DevOps Lab Cheat Sheet

A-1

Commands to demonstrate creating a project in local and remote repository using Git Bash and GitHub, applying `init`, `status`, `log`, `add`, `commit`, `push`, `config`, `clone`, and `reset` in Git`.

First time pushing code into a new repository

Step:1: git config --global user.name "YourName"

git config --global user.email "your_email@example.com"

Step 2: git init

Step 3: git status

Step 4: git add .

Step 5: git commit -m "Initial commit"

Step 6: git log

Step 7: git remote add origin https://github.com/username/repository-name.git

Step 8: git push -u origin master

Step 9: git clone https://github.com/username/repository-name.git

Step 10: git reset --soft HEAD~1

git reset --hard HEAD~1

If the repository already Exits

Step 1: git config --global user.name "YourName"

git config --global user.email your_email@example.com

Step 2: git clone https://github.com/username/repository-name.git

Step 3: cd repository-name

Step 4: git status

Step 5: git add .

Step 6: git commit -m "Updated commit"

Step 7: git log

Step 8: git push -u origin master

Step 9: git reset --soft HEAD~1

Step 10: git reset --hard HEAD~1

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Erros	Solution
remote origin already exists	git remote remove origin git remote add origin https://github.com/username/repo.git
Force Update Remote Repository Account	git remote set-url origin https://username@gitlab.com/username/project.git

A-2

Demonstrate how to create a project in a remote repository and apply fork, merge, diff, merge conflict, branch, and pull request concepts on the repository using GitHub.

Step1: Create Repository in GitHub

Open **GitHub**.
Click **New Repository**.
Enter repository name: **git-demo**

Step 2: Clone Repository to Local System

Open terminal and run:
git clone https://github.com/username/git-demo.git
cd git-demo

8 Step 3: Create a file and Write the content

Hello GitHub

Step 4: Commit and Push

git add file1.txt
git commit -m "Initial commit"
git push origin main

Step 5: Create and Switch to Branch

git branch feature1
git checkout feature1

Step 6: Modify File in Feature Branch

Open file1.txt and add a line
Now Commit changes
git add file1.txt
git commit -m "Feature branch update"
git push origin feature1

Step 7: Show Difference Between Branches

git diff main feature1

Step 8: Create Merge Conflict

git checkout main
Open file1.txt and add another line
git add file1.txt
git commit -m "Main branch change"
git merge feature1

Step 9: Resolve Conflict

Open file1.txt and add another line
git add file1.txt
git commit -m "Merge conflict resolved"
git push origin main

Step 10: Fork and Create Pull Request

Open repository in GitHub.
Click Fork.
Modify the file in forked repository.
Click Pull Request.
Click Create Pull Request to request merge into original repository.

A-3

Demonstrate the process of integrating a GitHub repository with Jenkins to automate project execution in a CI/CD pipeline.

Requisites: GitHub Repository with Java/Python files
Jenkins(Go to https://www.jenkins.io, Go to https://www.jenkins.io)

To Unlock Jenkins:

Open a browser and go to: http://localhost:8080
Jenkins will ask for the Administrator password.
Copy the password from: C:\Program data\Jenkins\secrets\initialAdminPassword
Install Suggested Plugins

I. Manual Build creation

In Jenkins:

Click New Item
Enter project name
Select Freestyle Project
Click OK

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Configure GitHub Repository

In Source Code Management

Select Git

Enter repository URL

Add Build Step

Scroll to Build → Add Build Step

Execute Windows Batch Command

Give the command (echo, python, java)

Save the Job

Run Build Now

II. Scheduled Build or Periodic Build (using Poll SCM)

In Jenkins:

Click New Item

Enter project name

Select Freestyle Project

Click OK

Configure GitHub Repository

In Source Code Management

Select Git

Enter repository URL

Enable Scheduled Build

Scroll to Build Trigger

Select Poll SCM

Mention * * * * *

Scroll to Build

Add Build Step → Execute Windows Batch Command

Give the command (echo, python, java)

Save

Make a change in the GitHub repository and the code.
Jenkins will detect the change using Poll SCM and automatically start the build.

III. Event-Based Automatic Build using GitHub Webhooks

In Jenkins:

Click New Item

Enter project name

Select Freestyle Project

Click OK

Configure GitHub Repository

Scroll to **Source Code Management**

Select **Git**

Enter the GitHub repository URL

Scroll to Build Triggers

Select: GitHub hook trigger for GITScm polling

Install and Run ngrok

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ngrok http 8080 (Jenkins port)

Add Build Step

Scroll to Build → Add Build Step
Execute Windows Batch Command
Give the command (echo, python, java)
Save the Job

Configure Webhook in GitHub

Open your repository in GitHub
Go to Settings
Click Webhooks
Click Add Webhook
Enter Webhook Details:
Payload URL: http://ngrok_generated_IP/**github-webhook/**
Click Add Webhook
Make a change in the project and push the code

When code is pushed to GitHub, the Webhook sends a request through ngrok to Jenkins, and Jenkins immediately starts the build.

B1

Create a Docker image for an application stored in a local repository and run the application using docker image.

Step1: Download Docker Desktop from the official website:

https://www.docker.com/products/docker-desktop
Install Docker Desktop.
Start Docker Desktop.
Verify installation in terminal / command prompt**: docker –version**

Step 2: Create a Local Application

Create a folder for the application**. Eg*: myapp***
Inside the folder create a simple Python application. ***App.py***

Step 3: Create a Dockerfile

Inside the same folder create a file named**: Dockerfile(**without .txt extension)
Add the following content:
FROM python:3.10
WORKDIR /app
COPY . .
CMD ["python","app.py"]

Step 4: Open terminal in the myapp folder.

docker build -t myapp .

Step 5: Verify the docker image

docker images

Step 6: Run the Docker Container

docker run myapp

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B2: Create and configure Jenkins files for workflow and build of an application and push the image on docker hub..

Step 1: Build Docker Image from Your Folder

cd your-folder-name

Build the image: docker build -t imagename .

Step 2: Add a Tag:

docker tag imagename yourdockerhubusername/imagename:tagname

eg: *docker tag myapp akshatha123/myapp:v1*

Step 3: Login to Docker Hub

Step 4: Push Image to Docker Hub

docker push yourdockerhubusername/imagename:tagname

eg: *docker push akshatha123/myapp:v1*

Step 5: Verify on Docker Hub

Step 6: Push the code to Github

git init

git add .

git commit -m "Initial commit"

git remote add origin https://github.com/your-username/docker_demo.git

git push -u origin main

Step 7: Add Docker Hub Credentials in Jenkins

Open Jenkins

Manage Jenkins → Credentials → Configure Credentials

Add:

Kind: Username & Password
ID: `dockerhub-creds`
Username: your Docker Hub username
Password: **Docker Hub Access Token**

How to Check Docker Access Token

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Step 1: Open Security Settings

Login to Docker Hub

Click your profile (top right)

Go to:

`Account Settings → Security`

Open **Access Tokens**

In command prompt:

```
echo YOUR_TOKEN | docker login -u your-username --password-stdin
```

Step 8: Create Jenkins Pipeline Job

Jenkins Dashboard → **New Item**
Select **Pipeline**
Configure:
Pipeline → *Pipeline script from SCM*
SCM → Git
Repo URL → your GitHub repo
Script Path → `Jenkinsfile`

Step 9: Build Now (manual)